

VRsense: An Explainable System to Help Mitigate Cybersickness in VR Games

Dinithi Dissanayake

Augmented Human Lab, School of Computing
National University of Singapore
Singapore, Singapore
dinithid@comp.nus.edu.sg

Chitralkha Gupta

Augmented Human Lab, School of Computing
National University of Singapore
Singapore, Singapore
chitralkha@nus.edu.sg

Prasanth Sasikumar

Augmented Human Lab, School of Computing
National University of Singapore
Singapore, Singapore, Singapore
prasanth.sasikumar@nus.edu.sg

Suranga Nanayakkara

Augmented Human Lab
School of Computing, National University of Singapore
Singapore, Singapore
suranga@ahlab.org



Figure 1: VRsense can detect cybersickness and provide insights into the reasons behind its predictions aiding VR game developers to make better design decisions

Abstract

Cybersickness remains a significant challenge in immersive VR gaming, affecting player comfort and engagement. In this study, we analyze data from approximately 400 players across 80 games to investigate the factors contributing to cybersickness in active gameplay environments. To bridge the gap between research and practical application, we present our late-breaking work, VRsense, an explainable system designed to assist VR game developers in evaluating cybersickness in their games. Unlike traditional black-box approaches, our system employs engineered features offering actionable insights into game design and user interactions. Feedback from eight VR professionals highlights the utility of the system and suggests avenues for adaptive gameplay strategies, such as optimizing task flows. Despite challenges in generalizability and subjectivity, our iterative approach lays the foundation for a deployable tool to assist developers in creating less cybersickness-inducing games.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719891>

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Participatory design*.

Keywords

VR Gaming, Cybersickness, Explainable Machine Learning

ACM Reference Format:

Dinithi Dissanayake, Chitralkha Gupta, Prasanth Sasikumar, and Suranga Nanayakkara. 2025. VRsense: An Explainable System to Help Mitigate Cybersickness in VR Games. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3719891>

1 Introduction

Virtual Reality (VR) is increasingly gaining prominence as a medium across various domains due to its ability to deliver highly immersive and satisfying user experiences. Among these domains, the gaming industry continues to be a primary driver of VR adoption, with VR gaming revenue projected to reach \$45.09 billion by 2024¹. This growth is driven by developers creating more immersive VR experiences [6, 18, 19], encouraging gamers to use VR headsets for greater engagement further leading to better VR hardware and gaming platforms.

¹<https://umatechnology.org/20-immersive-virtual-reality-statistics-to-know-in-2024/>

Despite these advancements, a significant challenge affecting the VR gaming experience is visually induced motion sickness (VIMS), commonly referred to as cybersickness [4, 21, 24, 25, 33]. Cybersickness can be understood through sensory conflict theories which suggest that symptoms such as dizziness and nausea arise from discrepancies between the movement perceived by the vestibular system and the visual input displayed on the VR screen [12, 14]. Cybersickness remains a significant barrier to the adoption of VR gaming, as it adversely impacts both the long-term gaming experience and the willingness of users to continue playing [24, 26].

Extensive research has been conducted to detect instances of cybersickness and evaluate the impact of various features on its occurrence. Previous studies have highlighted certain features as particularly impactful, often using controlled environments with isolated variables for user studies [5, 9, 27, 31, 35]. However, these environments may not accurately represent real-world gaming conditions, leading to unreliable or mismatched findings when applied to commercial games. Moreover, a substantial portion of the literature focuses on cybersickness induced passively while watching pre-recorded VR gameplay videos [11–13, 16]. This approach fails to account for the interactive nature of gaming, where user actions and direct engagement significantly influence the experience of cybersickness [21]. The causes of cybersickness may stem from the game’s design and interaction mechanisms, the player’s behavior, or even the individual’s inherent susceptibility to cybersickness. Thus, there is a critical need for comprehensive analysis of cybersickness experiences in active VR gameplay in real-world gaming scenarios.

Our aim is to enhance game designs and improve player experiences by providing actionable insights to VR game developers. To this end, we explore how players are affected by cybersickness in real-world gaming environments and present VRSense, a framework designed to detect cybersickness during gameplay and offer meaningful feedback for refining VR game environments. Therefore, through this study, we seek to address the following research questions:

- RQ1: What are the observed impact, and the key contributing factors of cybersickness on players during VR gameplay?
- RQ2: What is VR game developers’ perspective of cybersickness during game design?
- RQ3: How can explanatory systems for cybersickness detection support game developers in designing less cybersickness-inducing games?

This late-breaking work aims to bridge the gap between the players’ experiences of cybersickness during VR gameplay and the game developers’ needs for actionable, and explainable insights to design VR games that minimize cybersickness. Our framework, VRSense, integrates a cybersickness detection model with an explainability feature to provide insights into the model’s predictions. To evaluate its relevance and practical utility, we conducted a formative study with eight professional VR game developers. This study examines how such a system could integrate with and enhance the existing workflows of VR game developers. Based on the findings, we present insights to guide researchers in advancing cybersickness detection and mitigation research, emphasizing usability and practical applicability.

2 Related Works

2.1 Understanding and Mitigating cybersickness

Motion sickness occurs when there is a mismatch between actual and expected motion, leading to symptoms such as disorientation, headache, nausea, and eye strain [17]. In VR, motion sickness mostly referred to as cybersickness is a significant barrier to its adoption. Previous studies have explored various factors influencing cybersickness, including demographics such as age, gender, and prior VR experience [5, 8, 22, 23]. Research has also examined the effects of visual elements like field of view (FoV), depth of field (DoF) [31], scene content [9], illumination [2], and graphical details [27], as well as movement patterns like steering, jumping, and teleportation [35] and also virtual camera movements [9]. Techniques to reduce cybersickness include using reference objects [37], peripheral blurring [3], adjusting field of view [7], leveraging change blindness [32], applying visual guides [30] and reverse optical flow [24]. Most of these studies are conducted in controlled virtual environments that isolate specific factors [5, 9, 27, 31, 35]. To support real-world research, VR.NET [36] provides motion sickness ratings from actual VR gameplay enabling a broader understanding of cybersickness. This dataset contains recordings of VR gameplay of more than 400 participants along with their cybersickness labels. We are at the forefront of leveraging this dataset to investigate the occurrence and impact of cybersickness in real-world VR gaming scenarios to explore this phenomenon outside controlled experimental settings.

2.2 Modeling and Detection of cybersickness in VR

Deep learning (DL) and machine learning (ML) have recently gained popularity in cybersickness detection, employing models such as CNNs, LSTMs, DNNs, and attention-based transformers [10–13, 15, 16, 28, 34]. These models utilize various data modalities, including head-mounted display (HMD) data, gaze tracking, video, and physiological data, enabling multi-modal modeling in this domain. Building on this, we introduce a novel approach by incorporating game-interaction features, including controller usage, into cybersickness detection models. Unlike most DL-based models trained and tested on the same VR environments, our focus is on predicting cybersickness for unseen games—a more realistic and challenging use case. Our framework bridges the gap between model development and practical utility, guiding VR developers in evaluating and improving gaming environments. Most of the currently developed cybersickness detection models rely on black-box architectures, which lack explainability [16]. Incorporating explainability offers a valuable opportunity to enhance understanding and provide actionable insights for VR game developers. These insights empower developers to refine their designs and create user-friendly gaming experiences. Building on this concept, we present VRSense, a framework designed to assist VR developers in creating games that minimize cybersickness.

Explainability libraries like LIME [29] and SHAP [20] address a key gap in cybersickness detection by providing interpretable results. For example, LiteVR [16] used SHAP to extract global explanations and refine input features, enhancing model accuracy during design. Similarly, recent attention-based models have visualized

attention weights across features to provide global explanations of model behavior [12, 13]. Building on these advancements, our approach leverages LIME [29] for instance-level explanations. This allows us to identify why a specific segment of gameplay was predicted to induce cybersickness, offering actionable insights for both research and practical applications.

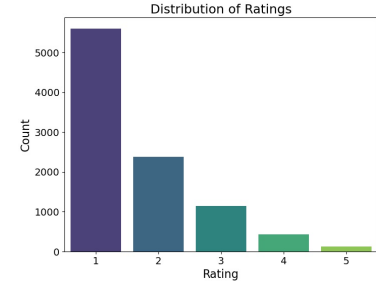
3 Cybersickness in Active VR Gameplay

3.1 Dataset - A Descriptive Analysis

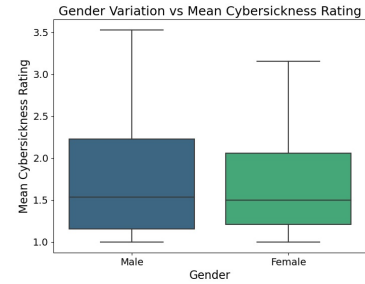
To support our analysis and achieve our objectives, we utilized the VR.NET dataset [36], a rich source of real-world gameplay data from individuals who reported their cybersickness levels using the Fast Motion Sickness Scale (FMSS). This scale, ranging from 1 (no symptoms) to 5 (extreme cybersickness), is widely adopted in recent literature for assessing cybersickness [1, 15]. Our study marks the first detailed analysis of this dataset, providing unique insights into how cybersickness impacts players during real-world active VR gameplay. The dataset comprises 161 hours of gameplay from 411 participants, each playing up to two sessions, resulting in a total of 793 sessions across 82 games on one of the most popular VR gaming platforms available. This pioneering analysis offers valuable findings and lays the groundwork for advancing cybersickness research in VR environments.

Participants were asked to play for 12 minutes, with the option to stop if they could not continue. Approximately 17% of the sessions were terminated by players due to severe cybersickness before completing the full 12 minutes. In 77% of the sessions, participants experienced some level of cybersickness, reporting a score of 2 or higher on the FMSS. Of these, 15% reported extreme cybersickness, with scores of 5. Notably, no participant who reported a score higher than 1 returned to a score of 1 during the 12 minutes of gameplay, indicating that once a player felt cybersick, they did not recover to a normal state within the session. The range of reported cybersickness scores across all the sessions can be seen in Figure 2(a). As playtime increases, the number of individuals experiencing cybersickness tends to increase. This trend highlights the significance of exposure duration as a key factor in understanding and discussing cybersickness.

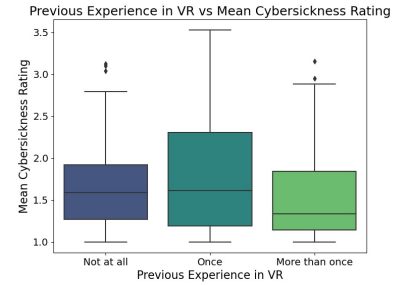
The dataset included demographic information, enabling the mapping of cybersickness ratings to demographic data for 200 participants. The participants ranged in age from 18 to 60 years, with 89% of them falling within the 18 to 40 age group. Of these, 99 were male, 97 were female, and 4 did not specify their gender. Using a Kruskal-Wallis H test (data was with a non-normal distribution), we found no significant difference in the mean(or max) cybersickness ratings between male and female participants. This challenges prior studies that often reported higher susceptibility to cybersickness in females, which may have been influenced by the under-representation of female participants in those studies as put forward by Macarthur et. al. [22]. We also examined the impact of prior VR experience, categorizing participants into three groups: no experience, experienced just once, and more than once. The median of the mean(and max) cybersickness rating of participants with prior VR experience was lower compared to the other two groups. However, a Kruskal-Wallis H test, suitable for non-normal data, revealed no significant differences in cybersickness ratings



(a)



(b)



(c)

Figure 2: (a) Distribution of cybersickness ratings across all participants, (b) Comparison of cybersickness ratings by gender, and (c) Analysis of cybersickness ratings based on participants' prior VR experience.

across these groups when considering the mean ratings. In contrast, when analyzing the maximum cybersickness ratings, the test indicated a significant difference between the groups. This suggests that while experienced VR users may still experience cybersickness during a 12-minute gameplay session, they are less likely to reach the maximum severity compared to less experienced users. The distributions for these analyses are illustrated in Figure 2(b) and Figure 2(c).

3.2 Game Genre vs Cybersickness

We aimed to explore how game genres influenced cybersickness and whether certain genres were more likely to induce higher levels of cybersickness. Through the platform tags we classified the

82 games to different genres including Adventure, Puzzle, Exploration, Hangout, PVP, PVE, Party Game, Racing, Sports, and Art. We explored the distribution of mean cybersickness ratings across the sessions and games, grouped by their respective genres. We can observe that puzzle games tend to have a higher median for cybersickness, suggesting that cognitive load may influence its occurrence. These games often include exploration features, such as searching for puzzle pieces, and text-based screens for instructions, which could contribute to the increased incidence of cybersickness. Interestingly, the art category reporting the highest median includes sandbox games featuring superhero or science fiction characters, where players often gain abilities like flying. We observed that participants who chose to fly instead of walk experienced higher levels of cybersickness. This underscores the critical role of user interactions and game design elements in the prevalence of cybersickness. The detailed distribution is provided in the Appendix.

4 Designing VRSense - an Explainable Cybersickness Detection Framework

Our research team consists of an interdisciplinary group of experts, including VR researchers, AI/ML researchers, and professionals from the VR gaming industry. Through several rounds of discussions, the team explored how a cybersickness detection model could enhance the quality of VR games by making them less likely to induce cybersickness. One key theme that emerged from these discussions was the importance of explaining why certain segments of gameplay are more likely to result in higher cybersickness. This highlighted the need for a system to provide explanations for its predictions. Therefore, we opted to use several machine learning models instead of deep learning techniques to prioritize explainability in the model's predictions. As the initial step in this iterative system design, we extracted features from the VR.NET dataset [36]. We then developed a baseline model for cybersickness detection and employed LIME to explain why specific timeframes were identified as inducing high levels of cybersickness.

4.1 Feature Extraction and Data Aggregation

We identified two controllable parameters critical to mitigating cybersickness: (1) game design elements and (2) user interactions. Game design elements fall under the purview of VR game developers, enabling them to make informed choices to design games that are less likely to induce cybersickness. In contrast, user interactions, while outside the direct control of developers, can be addressed by creating adaptive VR games to accommodate these limitations. Based on this understanding, we extracted the features listed in Table 1. To simplify the time-series nature of the data, we aggregated it into intervals by assigning the previous period's data to the current cybersickness rating. We tested three aggregation strategies—1-minute intervals, 2-minute intervals, and 3-minute intervals—and empirically found that the 3-minute intervals yielded the best performance.

4.2 Data Modeling and Explainability

Since the primary goal of our system is to evaluate new games, we structured our train/validation/test splits by games to ensure the model was tested on unseen games. Regarding the detection

variable, the Fast Motion Sickness Scale (FMSS) categories (1, 2, 3, 4, and 5) exhibited significant class imbalance, as shown in Figure 2(a) – therefore, we grouped the FMSS ratings into two categories: "not cybersick" (FMSS = 1) and "cybersick" (FMSS = 2, 3, 4, 5). To further mitigate the imbalance in the binary classification task, we implemented a class balancing mechanism by randomly under-sampling the overrepresented classes before training. This approach helped to create a more balanced dataset, improving model training and performance. As part of this exploratory work, we experimented with various machine learning models, with the Gradient Boosting model emerging as the most promising baseline. While its performance—60% accuracy and an F1 score of 0.67—indicates potential, we recognize that further refinements are necessary to improve these metrics. This result serves as a foundational step in our iterative process, paving the way for more sophisticated approaches in future iterations.

We integrated LIME for local explainability, allowing us to analyze individual predictions and determine how features contributed either positively (indicating cybersickness) or negatively (indicating no cybersickness). This instance-level analysis enabled us to extract key features with a high positive impact on cybersickness predictions, generating textual explanations such as "Changes in brightness levels are high," which provide actionable insights. The results from our explainability tools were presented in a demonstration to VR gaming professionals to gather feedback on the system's utility and alignment with industry needs. A detailed visualization of the current pipeline is provided in Figure 1.

5 Formative Study

With this context, we conducted a formative survey involving eight professionals from the VR gaming industry. The participants included six VR game developers, a 3D artist, and a VR product designer (aged between 25 and 40 years, with 5 to 10 years of industry experience). The genres of games developed by the participants ranged from action/adventure to simulation and puzzle games. All participants used Unity as their primary development platform, with one also utilizing Unity in combination with Unreal Engine. Their development experience spanned various VR headsets, including HTC Vive, Quest, and Vision Pro. The survey was conducted through a structured, web-based questionnaire, divided into three sections. The questionnaire is provided at this link.

Part 1: Current Perspectives on CyberSickness in VR Gaming - We explored how professionals in the VR gaming industry view cybersickness and its impact on game development. Specifically, we investigated whether they consider cybersickness a significant issue and the reasons behind their opinions. Participants were also asked about their current methods for assessing cybersickness in their games. Additionally, we gathered insights on the features they believe are most relevant for detecting cybersickness.

Part 2: Evaluating System Output and Usefulness - In this section, we provided participants with a sample video showing a gameplay session where the player reported high-cybersickness. We asked them to analyze the video and describe how they would address this feedback. Subsequently, they were shown the output from our framework, which included three reasons for the system's prediction of high cybersickness. Participants were then asked

Table 1: The resultant features employed in the model include the highlighted features. 'R' and 'L' denote the right and left sides, while 'y', 'p', and 'r' represent yaw, pitch, and roll, respectively.

Factor	Raw Features	Extracted Features
game design elements	camera position[x,y,z]	camera speed, velocity[x,y,z], acceleration[x,y,z], rotation[y,p,r]
	video frames	number of frames with text present, total change in brightness between frames
user interactions	hand and index triggers	total movements for index trigger[R,L], hand trigger[R,L]
	thumbstick movements	total movements for thumbstick[Rx,Ry,Lx,Ly]
	gaze pos[x,y,z],	eye speed, rotation[y,p,r],
	gaze rotation[x,y,z,w]	difference in head and eye directions[y,p,r]
	head position[x,y,z],	head speed, velocity[x,y,z], acceleration[x,y,z],
	head rotation[x,y,z,w]	rotation[yaw,pitch,roll]
	time(considered minute)	

how useful they found the system and its explanations for their development process.

Part 3: System Integration into Development Workflows - Finally, we focused on understanding how this system could integrate into the participants' existing workflows. We inquired about their current playtesting practices and the steps they take to evaluate their games. Participants were asked to reflect on how our system could fit into their workflows and the challenges they foresee in adopting such a system.

5.1 Key Findings

Mitigation of cybersickness is important: With the current frameworks, developers emphasized the importance of mitigating cybersickness in their work. One developer, primarily focused on professional development for research purposes, considered it only slightly important, as cybersickness was not a central concern for their projects. Three developers rated it as moderately important, citing reasons such as their target audience being less prone to cybersickness. Conversely, other developers stressed that mitigating cybersickness was very important. They highlighted that games must be enjoyable within the immersive worlds they create and noted the potential safety concerns when cybersickness is not adequately addressed. P6: *"It will ruin the entire experience the user has if they experience cybersickness during our simulations. They will be unable to immerse themselves within the world we made."*

Features for cybersickness detection: To refine features for cybersickness detection, we sought feedback from professionals, who identified camera settings, game mechanics, head movements, and game interactions as critical factors, while gaze patterns and environmental design were deemed less impactful. They emphasized technical aspects like FPS, FOV, and resolution. P2 remarked, *"The one thing and the most effective one, I believe, is the FPS. If the game drops frames and runs at a lower FPS, then, let's say 72, it affects cybersickness more than all other effectors."* They also highlighted scene movements and the balance between static and dynamic elements in the user's field of view.

System Integration: Participants rated the system's usefulness at 3.75/5 and its explanations at 4/5. While most professionals rated these aspects positively, one outlier provided a lower rating due

to the subjective nature of cybersickness. Professionals also noted they rely heavily on playtesting sessions, which varied in duration from 5 to 30 minutes, and user feedback for evaluation. Most preferred a standalone tool for post-playtesting analysis, with some suggesting integration into platforms like Unity or adding real-time feedback. Textual explanations were favored over visual formats like heatmaps, as they were seen as clearer and more actionable for game design improvements.

Challenges for the proposed system: We derived three main challenges from the qualitative feedback.

Generalizability: The professionals raised concerns about the generalizability of the system in two key areas. First, they questioned its ability to make predictions across diverse and dynamic games. P1: *"Motion sickness is heavily reliant on the gameplay that the game provides. So, having a system that can predict cyber sickness for a generic game would be tricky."* Second, they highlighted the subjective nature of player experiences, questioning whether results based on one player's gameplay could be generalized to others. P3: *"Every individual is different in terms of their perceived cybersickness so I can imagine several false positives or negatives unless it can be personalized or learn the behavior of the user."*

Validation of the Scores: Concerns about score validation were also raised, particularly regarding accuracy when data is limited. One participant commented, P4: *"There might be a situation where the system indicates users are undergoing intense cybersickness, while users' subjective feelings are not."*

Can all problems be fixed?: The professionals debated whether all cybersickness issues could be addressed, especially when these factors contribute to player enjoyment. P4 expressed, *"There might be some users who prefer the motion sickness adaptation, while others might feel a loss of control over the game."* To assist developers, P2 suggested creating a library of possible solutions tied to the reasons identified by the system.

6 Discussion, Limitations and Future Work

In this study, we investigated cybersickness in active gameplay environments, analyzing data from approximately 400 individuals across 80 games. Using a detailed descriptive analysis of the VR.NET dataset [36], we found minimal evidence of gender-based biases

in cybersickness susceptibility during gameplay. However, prior VR experience appeared to mitigate the severity of cybersickness. We also identified that extended exposure significantly increased sickness levels, establishing total exposure time as a critical variable for cybersickness studies. While exploring the relationship between game genres and cybersickness, the uneven data distribution across genres limited our conclusions. However, this analysis revealed that user interactions and game design elements play a significant role in influencing the incidence of cybersickness.

Drawing on a thorough literature review and iterative brainstorming, we identified key gaps between cybersickness detection research and its practical applications. To address these gaps, we developed VRsense - explainable machine learning (ML) framework capable of not only detecting cybersickness but also providing segment-specific explanations for sickness-inducing gameplay elements. Unlike deep learning (DL) approaches, which often function as "black boxes," we opted for ML due to its interpretability [16]. To improve the system's performance, we engineered several features, resulting in satisfactory accuracy and F1 scores. By incorporating a LIME-based model, we provided instance-level explanations, initiating an iterative design process. Feedback from professional VR game developers confirmed the system's utility, highlighting its potential for further enhancement.

However, as the formative study revealed, there is a potential concern regarding accuracy, as the current model achieved only 60%. This limitation stems from two main factors. First, our model predicts cybersickness in unseen games, which it has no prior knowledge of, and was trained on just 27 games[resultant subset of VR.Net after data cleaning]—insufficient for robust generalization. Additionally, class imbalance likely impacted the training dataset. Future efforts should focus on larger datasets with more cybersickness instances. Second, the model currently ignores the time-series nature of the data. We encourage the research community to consider these aspects—expanding datasets and leveraging temporal modeling—as future directions for advancing cybersickness detection models.

The study highlighted two primary contributors to cybersickness: game design elements and user interactions. Developers emphasized the importance of frame rate (FPS), resolution, and field of view (FOV), with FPS being cited as particularly impactful. These factors, while straightforward, are often hardware-dependent, limiting their variability in our dataset. Adaptive mechanisms, such as placing objects within the user's line of sight or guiding user focus with predetermined task flows, emerged as potential mitigation strategies. However, concerns were raised about whether all players would welcome such adaptations, as some might prefer an unaltered gameplay experience. As one developer suggested, "It'll be nice if [the system] gives players time to adapt and brings it back to the original state, or [allows] the player to choose themselves."

Professionals highlighted the issue of subjectivity emphasizing the variability in individual experiences, presenting challenges in the generalizability and validation of scores. To address this, we propose collecting more extensive data across diverse games and platforms, and refining the system to evaluate games using a cybersickness score derived from multiple gameplay sessions. Moving forward, we hope to incorporate these insights and further improve VRsense's accuracy while preserving its explainable nature. Next

we hope to enhance the pipeline by integrating a dashboard that enables developers to upload playtesting data and receive actionable feedback. Further collaboration with professionals will focus on embedding a refined prototype into their workflows, gathering insights to advance the system's practical utility.

Acknowledgments

We acknowledge the creation of the VR.NET dataset and the authors of the IEEE VR dataset paper, including Meta Reality Labs. Special thanks to Shreyas Sridhar for their support. Additionally, we appreciate the time and insights of the eight professional VR developers who participated in this study.

References

- [1] Samuel Ang and John Quarles. 2022. You're in for a bumpy ride! uneven terrain increases cybersickness while navigating with head mounted displays. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 428–435.
- [2] Muhammad Bahit, Sunu Wibirama, Hanung A Nugroho, Titis Wijayanto, and Mumtaz N Winadi. 2016. Investigation of visual attention in day-night driving simulator during cybersickness occurrence. In *2016 8th International Conference on Information Technology and Electrical Engineering (ICITEE)*. IEEE, 1–4.
- [3] Pulkit Budhiraja, Mark Roman Miller, Abhishek K Modi, and David Forsyth. 2017. Rotation blurring: use of artificial blurring to reduce cybersickness in virtual reality first person shooters. *arXiv preprint arXiv:1710.02599* (2017).
- [4] Eunhee Chang, Hyun Taek Kim, and Byoungyun Yoo. 2020. Virtual reality sickness: a review of causes and measurements. *International Journal of Human-Computer Interaction* 36, 17 (2020), 1658–1682.
- [5] Umer Asghar Chattha, Uzair Iqbal Janjua, Fozia Anwar, Tahir Mustafa Madni, Muhammad Faisal Cheema, and Sana Iqbal Janjua. 2020. Motion sickness in virtual reality: An empirical evaluation. *IEEE Access* 8 (2020), 130486–130499.
- [6] Sebastian Cmentowski, Andrey Krekhov, and Jens Krueger. 2019. Outstanding: a perspective-switching technique for covering large distances in VR games. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [7] Ajoy S Fernandes and Steven K Feiner. 2016. Combating VR sickness through subtle dynamic field-of-view modification. In *2016 IEEE symposium on 3D user interfaces (3DUI)*. IEEE, 201–210.
- [8] Guilherme Gonçalves, Miguel Melo, and Maximino Bessa. 2018. Virtual reality games: A study about the level of interaction vs. narrative and the gender in presence and cybersickness. In *2018 international conference on graphics and interaction (ICGI)*. IEEE, 1–8.
- [9] Ping Hu, Qi Sun, Piotr Didyk, Li-Yi Wei, and Arie E Kaufman. 2019. Reducing simulator sickness with perceptual camera control. *ACM Transactions on Graphics (TOG)* 38, 6 (2019), 1–12.
- [10] Rifatul Islam, Kevin Desai, and John Quarles. 2021. VR Sickness Prediction from Integrated HMD's Sensors using Multimodal Deep Fusion Network. *arXiv preprint arXiv:2108.06437* (2021).
- [11] Rifatul Islam, Kevin Desai, and John Quarles. 2022. Towards forecasting the onset of cybersickness by fusing physiological, head-tracking and eye-tracking with multimodal deep fusion network. In *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 121–130.
- [12] Dayoung Jeong and Kyungsik Han. 2024. PRECYSE: Predicting Cybersickness using Transformer for Multimodal Time-Series Sensor Data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 2 (2024), 1–24.
- [13] Dayoung Jeong, Seungwon Paik, YoungTae Noh, and Kyungsik Han. 2023. MAC: multimodal, attention-based cybersickness prediction modeling in virtual reality. *Virtual Reality* 27, 3 (2023), 2315–2330.
- [14] Dongyun Joo, Hanseob Kim, and Gerard Jounghyun Kim. 2024. The Effects of False but Stable Heart Rate Feedback on Cybersickness and User Experience in Virtual Reality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–15.
- [15] Behrang Keshavarz and Heiko Hecht. 2011. Validating an efficient method to quantify motion sickness. *Human factors* 53, 4 (2011), 415–426.
- [16] Ripan Kumar Kundu, Rifatul Islam, John Quarles, and Khaza Anuarul Hoque. 2023. LiteVR: Interpretable and Lightweight Cybersickness Detection using Explainable AI. In *2023 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. IEEE, 609–619.
- [17] Joseph J LaViola Jr. 2000. A discussion of cybersickness in virtual environments. *ACM Sigchi Bulletin* 32, 1 (2000), 47–56.
- [18] Jingyi Li, Filippo Frulli, Stella Clarke, and Andreas Butz. 2022. Towards Balancing Real-World Awareness and VR Immersion in Mobile VR. In *CHI Conference on*

- Human Factors in Computing Systems Extended Abstracts*. 1–6.
- [19] Xiaoxuan Li, Xiangshi Ren, Xin Suzuki, Naoaki Yamaji, Kin Wa Fung, and Yasuyuki Gondo. 2024. Designing a Multisensory VR Game Prototype for Older Adults-the Acceptability and Design Implications. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–18.
 - [20] Scott Lundberg. 2017. A unified approach to interpreting model predictions. *arXiv preprint arXiv:1705.07874* (2017).
 - [21] Tianren Luo, Fenglin Lu, Jiafu Lv, Xiaohui Tan, Chang Liu, Fangzhi Yan, Jin Huang, Chun Yu, Teng Han, and Feng Tian. 2024. Exploring Experience Gaps Between Active and Passive Users During Multi-user Locomotion in VR. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–19.
 - [22] Cayley MacArthur, Arielle Grinberg, Daniel Harley, and Mark Hancock. 2021. You're making me sick: A systematic review of how virtual reality research considers gender & cybersickness. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–15.
 - [23] Justin Munafo, Meg Diedrick, and Thomas A Stoffregen. 2017. The virtual reality head-mounted display Oculus Rift induces motion sickness and is sexist in its effects. *Experimental brain research* 235 (2017), 889–901.
 - [24] Su Han Park, Bin Han, and Gerard Jounghyun Kim. 2022. Mixing in reverse optical flow to mitigate vection and simulation sickness in virtual reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–11.
 - [25] Katharina Margareta Theresa Pöhlmann, Gang Li, Mark McGill, Reuben Markoff, and Stephen Anthony Brewster. 2023. You spin me right round, baby, right round: Examining the impact of multi-sensory self-motion cues on motion sickness during a VR reading task. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.
 - [26] Katharina Margareta Theresa Pöhlmann, Horia A Maior, Julia Föcker, Louise O'Hare, Adrian Parke, Aleksandra Ladowska, and Patrick Dickinson. 2023. I think I don't feel sick: Exploring the Relationship Between Cognitive Demand and Cybersickness in Virtual Reality using fNIRS. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.
 - [27] Matti Pouke, Arttu Tiiri, Steven M LaValle, and Timo Ojala. 2018. Effects of visual realism and moving detail on cybersickness. In *2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 665–666.
 - [28] Chenxin Qu, Xiaoping Che, Siqi Ma, and Shuqin Zhu. 2022. Bio-physiological-signals-based vr cybersickness detection. *CCF Transactions on Pervasive Computing and Interaction* 4, 3 (2022), 268–284.
 - [29] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. "Why should i trust you?" Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. 1135–1144.
 - [30] Kwang-Ho Seok, YeolHo Kim, Wookho Son, Yoon Sang Kim, et al. 2021. Using visual guides to reduce virtual reality sickness in first-person shooter games: Correlation analysis. *JMIR serious games* 9, 3 (2021), e18020.
 - [31] Rongkai Shi, Hai-Ning Liang, Yu Wu, Difeng Yu, and Wenge Xu. 2021. Virtual reality sickness mitigation methods: A comparative study in a racing game. *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 4, 1 (2021), 1–16.
 - [32] Evan A Suma, Seth Clark, David Krum, Samantha Finkelstein, Mark Bolas, and Zachary Warte. 2011. Leveraging change blindness for redirection in virtual environments. In *2011 IEEE Virtual Reality Conference*. IEEE, 159–166.
 - [33] Thomas Van Gemert, Niels Christian Nilsson, Teresa Hirzle, and Joanna Bergström. 2024. Sickknifical Steps: A Systematic Review and Meta-analysis of VR Sickness in Walking-based Locomotion for Virtual Reality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–36.
 - [34] Jialin Wang, Hai-Ning Liang, Diego Monteiro, Wenge Xu, and Jimin Xiao. 2022. Real-time prediction of simulator sickness in virtual reality games. *IEEE Transactions on Games* 15, 2 (2022), 252–261.
 - [35] Tim Weißker, André Kunert, Bernd Fröhlich, and Alexander Kulik. 2018. Spatial updating and simulator sickness during steering and jumping in immersive virtual environments. In *2018 IEEE conference on virtual reality and 3D user interfaces (VR)*. IEEE, 97–104.
 - [36] Elliott Wen, Chitraksha Gupta, Prasanth Sasikumar, Mark Billinghurst, James Wilmott, Emily Skow, Arindam Dey, and Suranga Nanayakkara. 2024. VR. net: A real-world dataset for virtual reality motion sickness research. *IEEE Transactions on Visualization and Computer Graphics* (2024).
 - [37] Carolin Wienrich, Christine Katharina Weidner, Celina Schatto, David Obremski, and Johann Habakuk Israel. 2018. A virtual nose as a rest-frame-the impact on simulator sickness and game experience. In *2018 10th international conference on virtual worlds and games for serious applications (VS-Games)*. IEEE, 1–8.

7 Appendices

A Genre vs Mean Cybersickness Rating

The genres we identified through the platform tags include Adventure (28), PVP (19), Hangout (11), Party Game (7), Exploration (4), Art (3), Puzzle (3), Sports (3), PVE (2), and Racing (2). Notably, the number of games belonging to the genres are not equally distributed, which skews our results. However, the medians of Puzzle and Art games are particularly interesting, as they appear to be higher than expected given their smaller sample sizes.

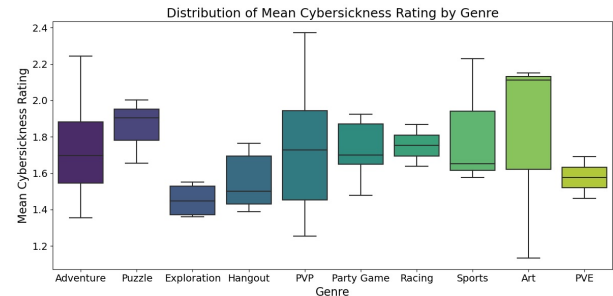


Figure 3: Genre vs Mean Cybersickness Rating